

TCP Flags — The Language of Every Scan

SYN

"I want to connect"

ACK

"I received your packet"

RST

"Close immediately"

FIN

"Done, close gracefully"

PSH

"Deliver data now"

URG

"Process immediately"

NULL

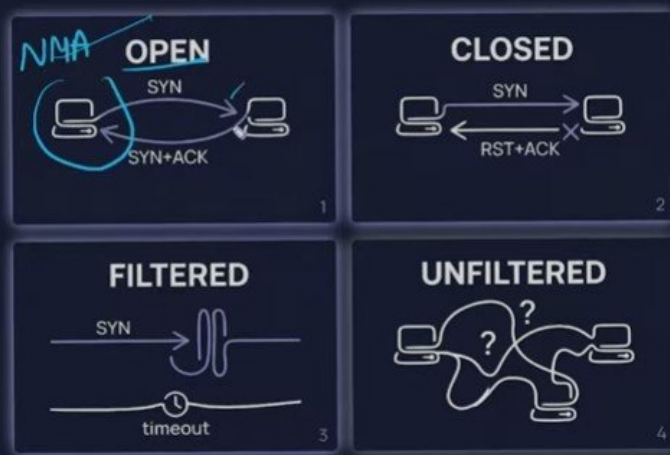
No flags — stealth



Mnemonic: SYN = Says Yes Now · ACK = Acknowledged · RST = Reject Suddenly · FIN = Finished



Port States — What NMAP Sees

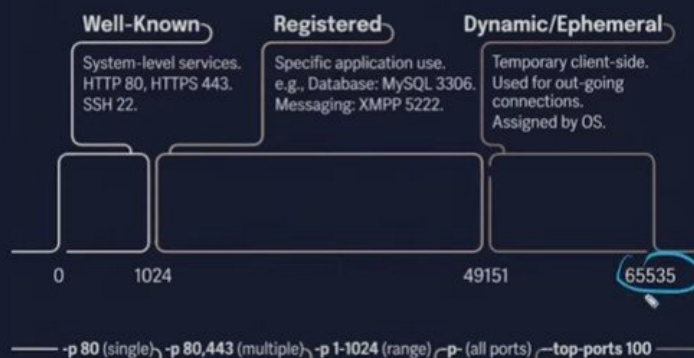


How NMAP Interprets Responses

- **Open:** SYN → SYN+ACK returned. App is listening.
- **Closed:** SYN → RST+ACK returned. Host up, no app.
- **Filtered:** SYN → no response (dropped). Firewall active.
- **Unfiltered:** Reachable, but state indeterminate.
- **open|filtered / closed|filtered:** Ambiguous — no conclusive response received.



Port Specification — Telling NMAP Which Ports to Scan



Port Range Syntax

Syntax	Example	Meaning
Single	<code>-p 80</code>	Port 80 only
Multiple	<code>-p 80,443,8080</code>	Specific ports
Range	<code>-p 1-1024</code>	Ports 1-1024
All	<code>-p-</code>	All 65,535 ports
Top N	<code>--top-ports 100</code>	Most common 100

Default (no `-p`): scans the **top 1,000** most common ports.



Host Discovery — Finding Live Targets

Key Flags

- -sn — Discover hosts only, skip port scan
- -Pn — Skip discovery, assume host is up (use when ICMP is blocked)

Always run host discovery before port scanning to avoid wasting time on dead hosts.



-sS → TCP SYN
 -sT → TCP connect scan

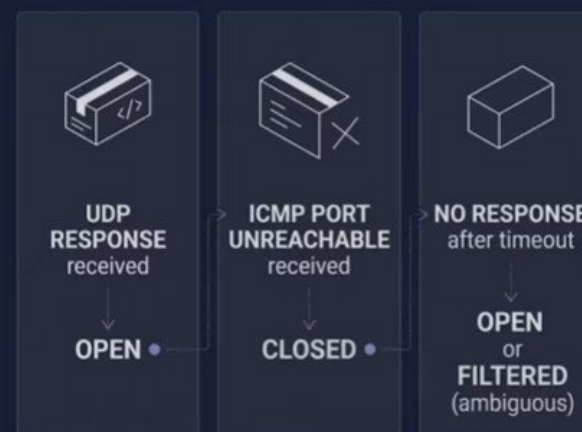
-sS(TCP SYN Scan) vs -sT(TCP Connect scan)

Feature	-sS	-sT
Root required	✓	x
Full handshake	x	✓
Speed	Faster	Slower



UDP Scan — The Slowest Scan

- UDP has no handshake (no connection setup like TCP)
- No confirmation (ACK), so no clear response from target
- Nmap has to wait for timeout to decide result
- Linux systems limit ICMP replies, so responses are delayed
- Result → UDP scan is very slow



Common UDP services: DNS port 53, DHCP ports 67-68, SNMP port 161, NTP port 123.

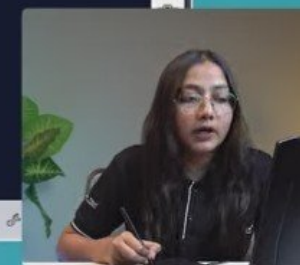
```
nmap -sU --top-ports 100 192.168.1.1
```



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UDP Scan



NULL, FIN & Xmas Scans

A closed port *must* reply with RST to any packet lacking SYN/ACK. NMAP exploits this. No response means OPEN or FILTERED.

Scan Type	Flags Sent	Open Response	Closed Response
FIN	FIN	No response	RST
NULL	None	No response	RST
<u>XMAS</u>	<u>FIN+PSH+URG</u>	No response	RST



ACK Scan — Mapping Firewall State (-sA)

ACK scan does **not** detect open ports. It reveals whether a firewall is **stateful** or **stateless** by sending a bare ACK packet and observing the response.

- Uses ACK flag to send packets (not SYN)
- Does not detect open/closed ports
- Helps identify firewall rules
- RST response → Unfiltered, No response → Filtered
- Used for network mapping & firewall detection

```
nmap -sA 192.168.1.0/24
```



NMAP Flags — Scan Types & Discovery

Scan Type Flags

Flag	Scan Type	Root?
-sS	SYN Stealth	✓
-sT	TCP Connect	X
-sU	UDP Scan	✓
-sN	NULL Scan	✓
-sF	FIN Scan	✓
-sX	Xmas Scan	✓
-sA	ACK Scan	✓
-sn	Ping Scan	X

Discovery Flags

Flag	Purpose
-Pn	Skip host discovery
-PS	TCP SYN ping
-PA	TCP ACK ping
-PE	ICMP Echo ping
-PP	ICMP Timestamp



OS Detection — TCP/IP Fingerprinting

- Sends specially crafted packets.
- Analyzes how the system responds.
- Checks details like:
 - TTL → How far packet travels
 - TCP Window Size → Data handling pattern
 - IPID Pattern → Packet ID sequence
- Compares these responses with its fingerprint database

Commands

nmap -O 192.168.1.1 — OS detection only

nmap -A 192.168.1.1 — OS + version + scripts + traceroute

to -D



Timing Templates



Timing (-T0 to -T5)

-T0 = max stealth (1 pkt/5 min); -T3 = default; -T5 = max speed, instantly detected

Levels:

- T0 (Paranoid): Very slow, maximum stealth
- T1 (Sneaky): Slow, avoids detection
- T2 (Polite): Low network load
- T3 (Normal): Default, balanced
- T4 (Aggressive): Fast, less stealth
- T5 (Insane): Very fast, less accurate



Fragment Packets (-f)



Splits packets into tiny fragments. Basic firewalls inspect only the first fragment, letting malicious payloads slip through undetected.

```
nmap -sS -f 192.168.1.1
nmap -sS --mtu 24 192.168.1.1
```

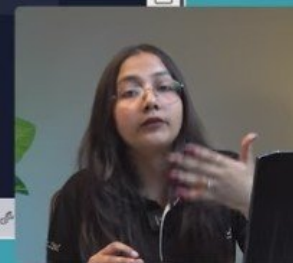
🍕 Like slicing a pizza — the guard checks only the first slice and waves the rest through.



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35°C Sunny



Advanced Evasion Techniques



Idle / Zombie Scan (-sI)

Uses a third-party zombie host. Your IP never touches the target.

```
nmap -sI zombie_host:port
target_ip
```



Source Port Manipulation

Spoofs source port. Firewalls often trust port 53 (DNS) blindly.

```
nmap -sS --source-port 53
192.168.1.1
```



Randomize Hosts

Scans subnet IPs randomly, breaking sequential scan detection.

```
nmap --randomize-hosts
192.168.1.0/24
```



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Earnings upcoming

NMAP Flags — Scan Types & Discovery

Scan Type Flags

Flag	Scan Type	Root?
-sS	SYN Stealth	✓
-sT	TCP Connect	X
-sU	UDP Scan	✓
-sN	NULL Scan	✓
-sF	FIN Scan	✓
-sX	Xmas Scan	✓
-sA	ACK Scan	✓
-sn	Ping Scan	X

Discovery Flags

Flag	Purpose
-Pn	Skip host discovery
-PS	TCP SYN ping
-PA	TCP ACK ping
-PE	ICMP Echo ping
-PP	ICMP Timestamp



Timing, Output & Advanced Flags

Timing Profiles

Flag	Profile	Speed
-T0	Paranoid	Extremely slow
-T1	Sneaky	Very slow
-T2	Polite	Slow
-T3	Normal	Default
-T4	Aggressive	Fast
-T5	Insane	Very fast

Output & Advanced

Flag	Format / Purpose
-oN	Normal text
-oX	XML
-oG	Grepable
-v	Verbose output
-A	OS + Version + Scripts + Traceroute
-O	OS fingerprinting
-sV	Service/version detection
--script	Run NSE scripts

☐ `nmap -sS -p 1-1000 -T4 -A 192.168.1.1` — SYN stealth scan, top 1000 ports, aggressive timing, full enumeration.



kali-linux-2025.4-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library

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My Computer

- kali-linux-2025.4-vmware-amd64
- Metasploitable2-Linux
- kali-linux-2026.1-vmware-amd64
- Debian 11.x 64-bit

Home kali-linux-2025.4-vmware-amd64 Metasploitable2-Linux

Session Actions Edit View Help

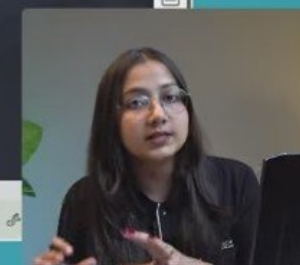
root@kali: /home/kali

```
(root@kali)~[/home/kali]
nmap -sT -p- -A -v 192.168.70.129 -oX nmapscan.xml
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-14 03:35 EDT
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 03:35
Completed NSE at 03:35, 0.00s elapsed
Initiating NSE at 03:35
Completed NSE at 03:35, 0.00s elapsed
Initiating NSE at 03:35
Completed NSE at 03:35, 0.00s elapsed
Initiating ARP Ping Scan at 03:35
Scanning 192.168.70.129 [1 port]
Completed ARP Ping Scan at 03:35, 0.08s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 03:35
Completed Parallel DNS resolution of 1 host. at 03:35, 0.06s elapsed
Initiating Connect Scan at 03:35
Scanning 192.168.70.129 [65535 ports]
Discovered open port 80/tcp on 192.168.70.129
Discovered open port 111/tcp on 192.168.70.129
Discovered open port 23/tcp on 192.168.70.129
Discovered open port 5900/tcp on 192.168.70.129
Discovered open port 25/tcp on 192.168.70.129
Discovered open port 22/tcp on 192.168.70.129
Discovered open port 21/tcp on 192.168.70.129
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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```

kali-linux-2025.4-vmware-amd64 - VMware Workstation
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My Computer
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Metasploitable2-Linux
kali-linux-2026.1-vmware-amd64
Debian 11.x 64-bit

kali-linux-2025.4-vmware-amd64
Session Actions Edit View Help
root@kali: /home/kali
root@kali: /home/kali

Read data files from: /usr/share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.40 seconds
Raw packets sent: 20 (1.626KB) | Rcvd: 16 (1.338KB)

(root@kali)~/home/kali
ls
1.c Desktop mitm newdir Pictures scan.xml text.txt
2.c Documents mtu new.tar Public spiderfoot Videos
archive1.tar.gz Downloads Music newzip.zip PyPhisher subfinder wscube.txt
archive.tar.gz hash.txt n0kovo_subdomains nmapscan.xml scan.html task.txt
armitage-tmp hash.xml newarchive.tar password.txt scan.txt Templates

(root@kali)~/home/kali
xsltproc nmapscan.xml -o nmapscan.html

(root@kali)~/home/kali
ls
1.c Desktop mitm newdir password.txt scan.txt Templates
2.c Documents mtu new.tar Pictures scan.xml text.txt
archive1.tar.gz Downloads Music newzip.zip Public spiderfoot Videos
archive.tar.gz hash.txt n0kovo_subdomains nmapscan.html PyPhisher subfinder wscube.t
armitage-tmp hash.xml newarchive.tar nmapscan.xml scan.html task.txt

(root@kali)~/home/kali

```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.



Key Concepts Recap



TCP Flags

SYN, ACK, RST, FIN, PSH, URG — the building blocks of every scan type.



Scan Types

From basic -sT to stealth -sI to evasive -sX — match the technique to the scenario.



Output Formats

Normal, Greppable, XML — always use -oA to capture all three at once.



Port States

Open, closed, filtered, unfiltered — each tells a distinct story about the target.



Service & OS Detection

-sV, -O, and NSE scripting reveal what's running beneath the surface.



Ethics & Legal

Never scan without explicit authorization. The law applies everywhere, just at work.



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Hot weather

